Forensic Microscopy

Lab 10: Hairs

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Where can you find the slides?

- https://aritra-evolves.github.io/FIVS215.html
- Canvas

Grab a microscope, set up kohler

Agenda

- Unskip
- Characteristics
- Identification
- Practice
- Course evaluation

Coming Up

Next Week: Open Lab/Honors Lab/Practice practical

Two Weeks: Lab Practical

- Make Your Lab Notebook ASAP
- See Week 1 Slides for Details
    - These will be the only type of notebook allowed in!

Post Labs:

Hair Identification due the week of Open Lab

- Everything will be graded before the practical so I need these in ASAP

Please don't forget to fill out the course evaluation when it's open

Characteristics

- Cuticle
  - i.e. Imbricate for Humans
- Medulla
  - Thickness, Transparency, Pattern
- Cortex
- Scales
- Distribution of Pigmentation
- Diameter
  - Human Avg. 80 micrometers
- Cannot be tied to a source with microscopy alone
- Not identical across sources

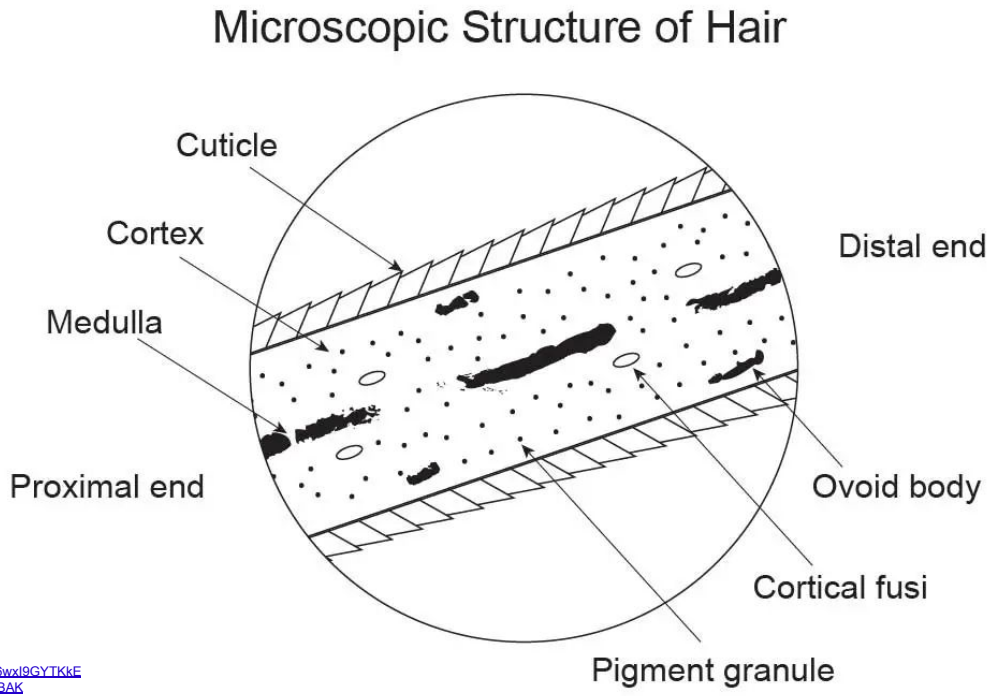


Image:
https://www.google.com/url?sa=i&url=https%3A%2F%2Frsscience.com%2Fhair-under-a-microscope%2F&psig=AOvVaw2WPn1ox6wxl9GYTKkEntoJ&ust=1763066386713000&source=images&cd=vfe&opi=89978449&ved=0CBYQjRxqFwoTCNi2mJ-87ZADFQAAAAAdAAAAABAK

Features of hair

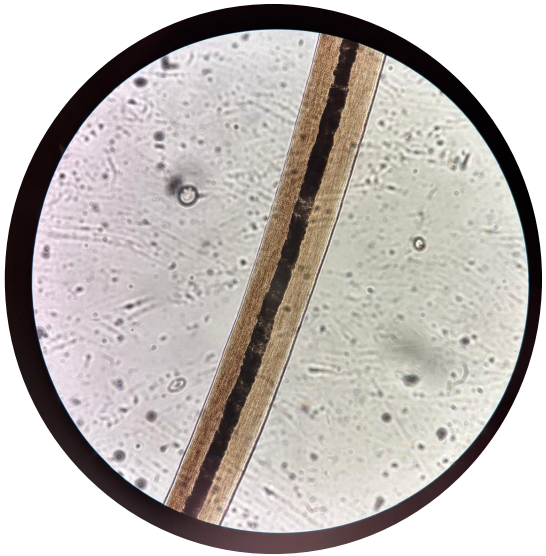


Anagen hair (10x)
(sheath, above the root)

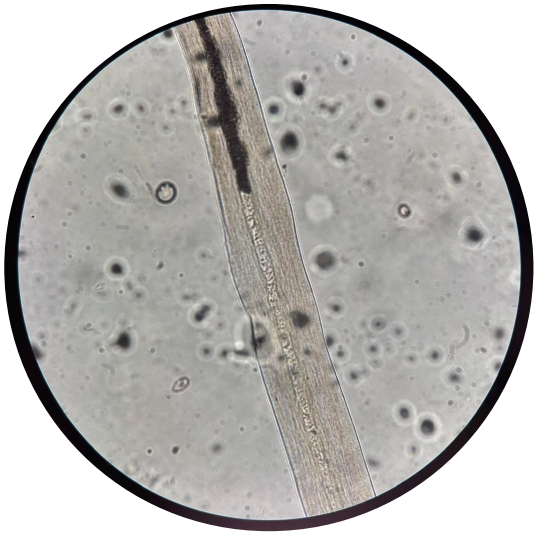


Telogen hair (20x)
(no sheath, club shaped root)

Features of hair

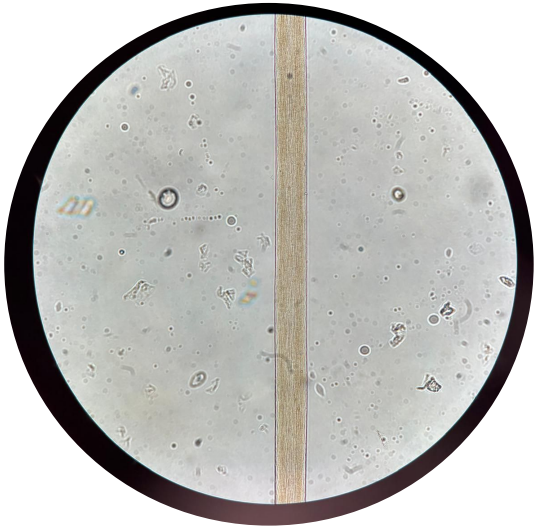


Scalp hair (20x)
(consistent thickness)

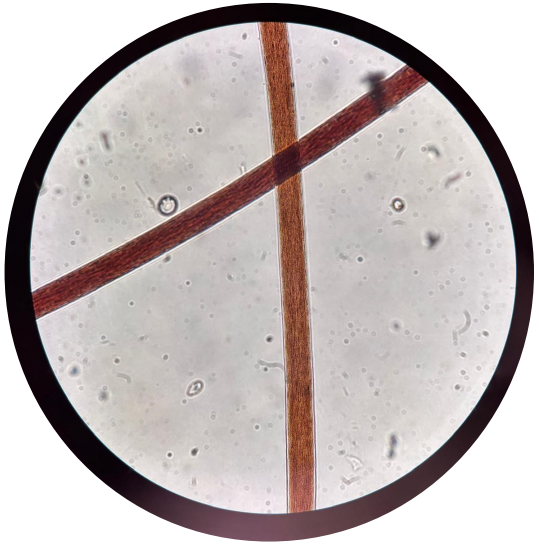


Pubic hair (20x)
(thickness changes)

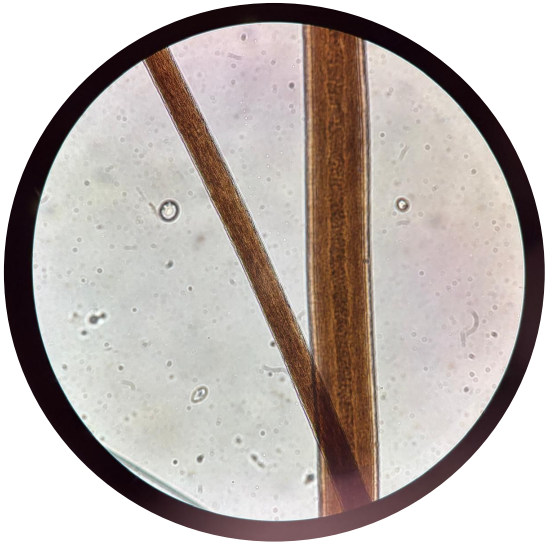
Features of hair



Light hair color (20x)

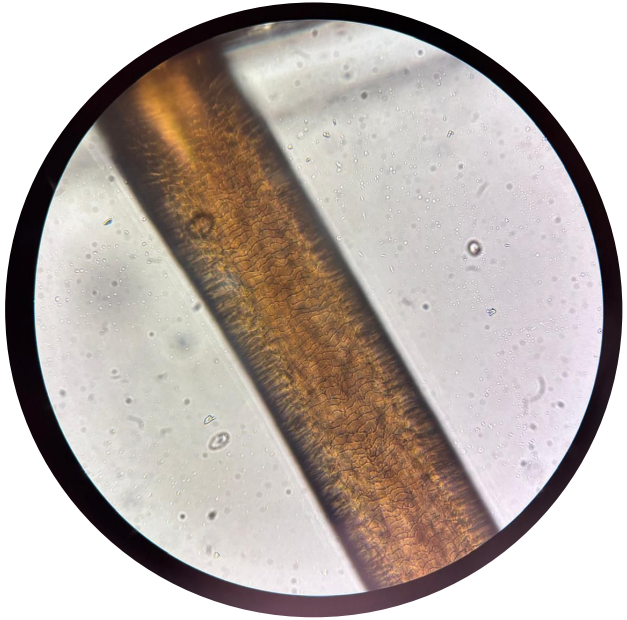


Medium hair color (20x)
(artificially dyed?)

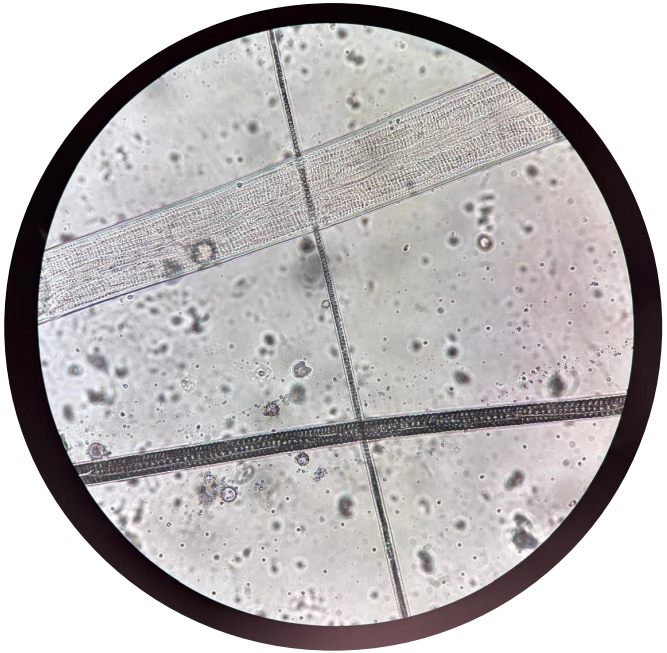


dark hair color (10x)

Features of hairs

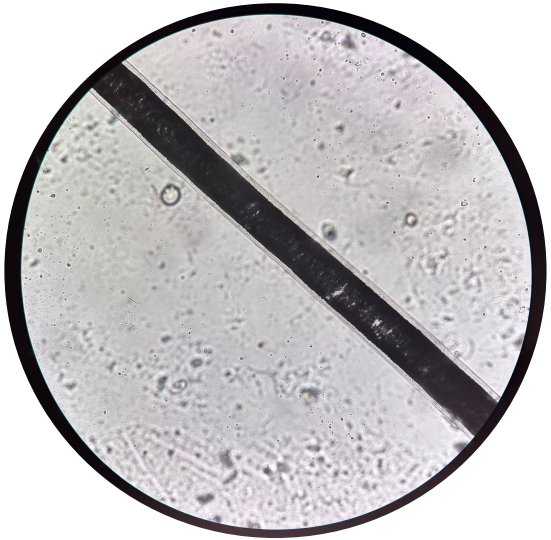


Cuticle pattern of human hair (40X)



Cuticle pattern of animal (arctic hare) hair (20X)

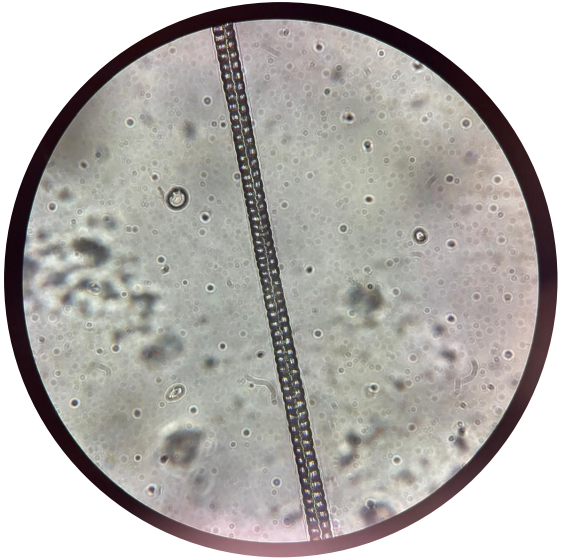
Features of hair



Cow hair (20X)



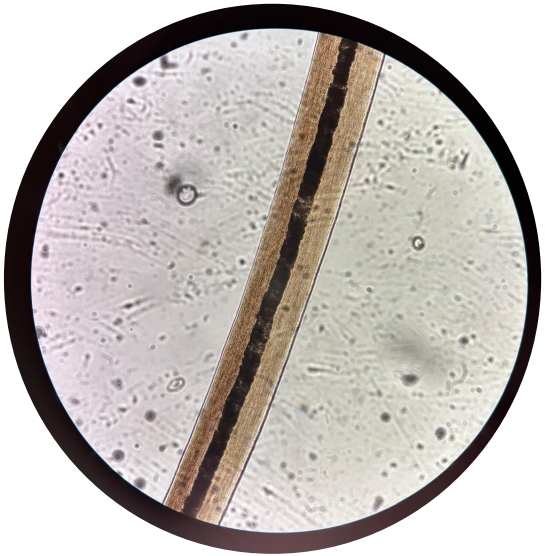
Yak hair (40X)



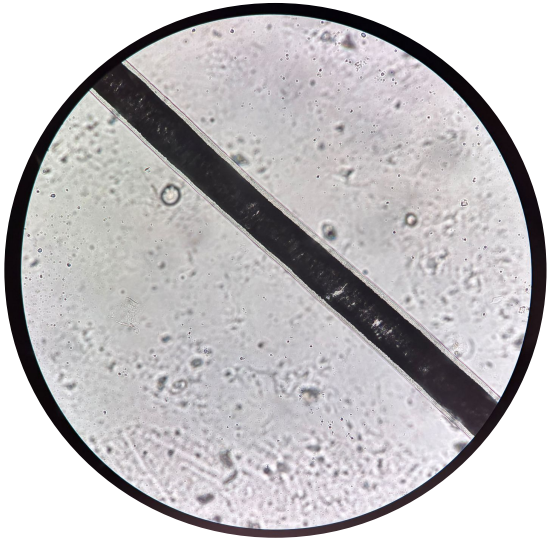
Arctic hare hair (40X)

Human vs Animal hair

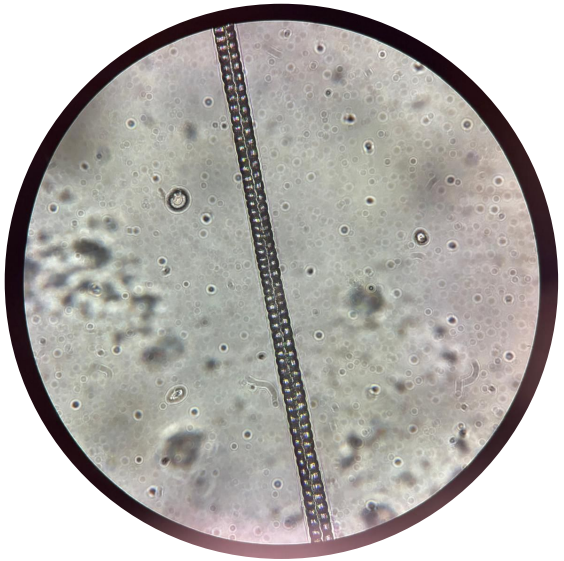
Medullary index: Medulla width/Hair Diameter



Human hair (20x)



Cow hair (20X)



Arctic hare hair (40X)

Animal hair: thicker and often patterned medulla

Identification

- Proximal vs. Distal End
- Somatic Origin
  - Width of Root
  - Coarseness
- Racial Origin
  - Using the Chart in the Protocol

Race	Diameter	X section	Pigmentation	Cuticle	Undulation
Negroid	60-90um	flat	Dense and clumped		prevalent
Caucasoid	70-100um	oval	Evenly distributed	mediu m	uncommon
Mongoloid	90-100um	round	Dense auburn	thick	never

Practice

1. Examine Animal Hairs
    a. Look at Three or More Samples
2. Examine Human Scalp Hair
    a. Look at Yours and a Partner's from Last Week
3. Determine Racial Origin
    a. Look at Two or More Samples
4. Determine Somatic Origin
    a. Look at Two or More Samples



Return from the Ball: Henry Gervex (1879)

Open lab is next week!